IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 5 February 2025, accepted 11 February 2025, date of publication 14 February 2025, date of current version 21 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3542284

RESEARCH ARTICLE

Optimizing Crop Recommendations With Improved Deep Belief Networks: A Multimodal Approach

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ABSTRACT With the increase in world population, agricultural planning is significant to ensure food security. Timely recommendations for crops could be valuable for planning food production and maintaining food sustainability. This proposed work suggests a crop recommendation model considering physical soil characteristics, chemical soil characteristics, climate, and crop characteristics, using Improved Deep Belief Networks (IDBN). For this study, four important Indian crops—rice, maize, finger millet and sugarcane —were taken into account. The soil, weather, and crop datasets were gathered from different sources and subjected to feature selections method to extract optimal features. The crop recommendation model was developed using the proposed IDBN with Gaussian Restricted Boltzmann Machines and Ranger Optimizer with selected optimal features as input. Gaussian Restricted Boltzmann Machines enable the use of continuous values, and Ranger Optimizer ensures rapid convergence with selected optimal features as input. The findings from the investigation confirm that the suggested IDBN outperforms conventional Deep Belief Networks (DBN). The effectiveness of the crop recommendation model was evaluated using IDBN, and its performance was compared with several other machine learning algorithms.

INDEX TERMS Deep belief networks, restricted Boltzmann machines, crop recommendation, DBN.

I. INTRODUCTION

Crop recommendations based on land diversity, soil mineral components, and climate characteristics are necessary to increase crop yields and improve productivity. Crop productivity is susceptible to climate change and not only affects crop yields but also has a significant impact on natural resources, such as land and water, vital for final crop yield [1]. As climate change continues to challenge agricultural production, suitable agricultural practices are necessary [2]. Suitable soil is the production medium for 95% of crops worldwide and provides structure and nutrients to crops [3], [4]. This calls for choosing crops that suit soil characteristics, thus ensuring sustainability [5]. Water shortages, land degradation, poor input utilization, and low productivity are characteristics of India's rainfed agriculture, which makes up around 54% of the country's total area. Hence, it is imperative to develop crop recommendation systems that consider weather and soil characteristics to help farmers choose suitable crops and improve their overall crop production. In the era of urbanization, agricultural areas are unlikely to increase; hence, this justifies the best use of available agricultural land to gain maximum yield. Crop recommendation is a challenging task because the number of features that should be considered is not established; the association between climate, soil, and crop characteristics is nonlinear; and machine learning is a promising approach for developing a crop recommendation model [6].

The associate editor coordinating the review of this manuscript and approving it for publication was Mostafa M. Fouda.

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Machine Learning (ML) is used to associate patterns based on training principles and has prominent forecasting, prediction, and recommendation capacities. Currently, ML algorithms are extensively leveraged in the field of agriculture to meet increasing food demand [7]. ML models are utilized for the functionalities such as crop yield prediction [8], fertilizer recommendation, weed detection [9], water management [10], and soil management [11]. Massive amounts of farming data are generated from soil sensor devices, satellites, and weather stations. The historical data collected to fathom crop yield, field operations, and fertilizer requirements are extensive. Agricultural big data require suitable data handling and processing platforms to analyze large-scale data generated and extract valuable information from it [12], [13], [14]. In recent times, as the availability of agricultural data numerous ML algorithms have been developed to predict agricultural variables, with neural networks being the most widely used due to their superior ability to accurately capture empirical relationships between features [15], [16]. However, traditional neural networks, with only one or two hidden layers, often lack the capacity to represent complex functions, limiting their ability to effectively learn from such intricate datasets. Deep Learning, a machine learning approach that extends traditional neural networks with multiple layers of hierarchical representation to model complex functions, has seen increasing application in agriculture in recent years [17]. Deep belief networks (DBN) are models with deep nonlinear structures and stacked Restricted Boltzmann Machines (RBMs) that can establish favorable relationships in the data. When the training data are limited, the DBN can be advantageous because it allows for weight fine-tuning using back propagation or other algorithms following the learning of initial weights via prior learning. The merits of DBNs are considered and used in agricultural applications [18], [19], [20]. The proposed models for effective crop recommendation are limited to the available literature [21]. The suggested crop recommendation methodology is therefore predicated on Improved Deep Belief Networks which recommends An appropriate crop for a location based on the soil, crop, and weather parameters. The irrigation and weighing lysimeter data for winter wheat and summer maize rotation system are observed in [22]. The structure of the paper is as follows: Part II discusses the background work relevant to the proposed model. part III proposes a crop recommendation framework established on improved deep belief networks. part IV shows the dataset used, implementation details of the proposed research work. Part V Discussion on experimental results. Part VI presents conclusions and suggestions for future research.

II. BACKGROUND WORK

Developments in artificial intelligence and precision agriculture have the potential to assess and plan agricultural management from available data. Addressing the challenges of agriculture requires a better understanding of complex, multivariate, and unpredictable agro ecosystems. This understanding is achieved by continuously measuring, monitoring, and analyzing various physical aspects and climatic occurrences. The amalgamation of machine-learning models has enhanced forecasting and prediction applications in the agricultural sector [21]. Research has investigated the use of machine learning and Internet of Things (IoT) technologies for crop recommendation, utilizing data collected from sensors and weather parameters. There are some of the most dangerous diseases in crops, such as late blight and early blight of potato, tomato leaf mold, and mango anthracnose, which could cause big losses when diagnosis and control are not timely [23]. Existing models typically incorporate soil type, temperature, and rainfall as training features, employing methods such as Support Vector Machines (SVMs) and Decision Trees for recommendations. However, these earlier studies often relied on a limited number of features and smaller datasets [24], [25]. Artificial Neural Networks (ANN) are applied to predict soybean harvest area, yield, and production, and the model was found to be more reliable than time-series prediction [26]. Reference [27] Proposed a crop recommendation model based on an ANN of soil, crop, and weather characteristics using a back propagation algorithm. An ANN was used to study the effects of variations in meteorological conditions on the concentrations of deoxynivalenol, nivalenol, and ferulic acid in wheat grains. Despite its popularity, an ANN trained with the back propagation algorithm is extremely subject to getting stuck in the local minima. To overcome this limitation, Deep Learning(DL) architectures can be used to construct feature layers to model the statistical structure of the input [28], [29]. DL architectures like Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have been introduced for crop yield prediction, yielding more favorable results compared to traditional machine learning algorithms [30]. Weed growth was analyzed using CNN, and different CNN architectures were compared. This study detected weed growth more accurately than in previous studies [31]. Restricted Boltzmann Machines are stacked to create Deep Belief Networks (DBNs), a form of deep neural network and probabilistic model that have demonstrated effectiveness in enhancing forecasting accuracy. The performance of the DBN was compared with other machine learning models, demonstrating superior results. References [32] and [33]. The DBN model was used for short-term drought prediction and was compared with the back propagation neural network. It was discovered that the DBN model was more accurate at predicting drought [34]. Proposed a computer vision model to detect leaf diseases and improve the yield. DBN showed improved classification accuracy compared to back propagation neural networks and recurrent neural networks in leaf disease detection. Each study considered the advantages of Deep Belief Networks [DBN] and established favorable relationships between features, thus increasing classification accuracy [35].

This study introduces a crop recommendation model utilizing Deep Belief Networks (DBNs) [36], which considers soil

and climate characteristics specific to Karnataka State, India. The research concentrates on four primary crops cultivated in the region: rice, sugarcane, maize, and finger millet. To enhance model efficiency, an adaptive learning step is proposed that selects the optimal learning rate after each iteration, enabling faster training. The model is further refined by integrating an improved Deep Belief Network with a Gaussian Bernoulli Restricted Boltzmann Machine [33] and a Ranger Optimizer for more accurate crop recommendations. Section III elaborates on the proposed method.

III. PROPOSED CROP RECOMMENDATION FRAMEWORK WITH IMPROVED DEEP BELIEF NETWORKS

The proposed crop recommendation model is developed using Improved Deep Belief Networks, which capture the dependencies of crops on soil and weather conditions. Weather data, collected from 1972 to 2017, is used in the model. The aim of the proposed work is to identify the relationship between location-specific soil characteristics and weather parameters to recommend the most suitable crop for each location. The proposed efficient crop recommendation model is shown in Figure 1.

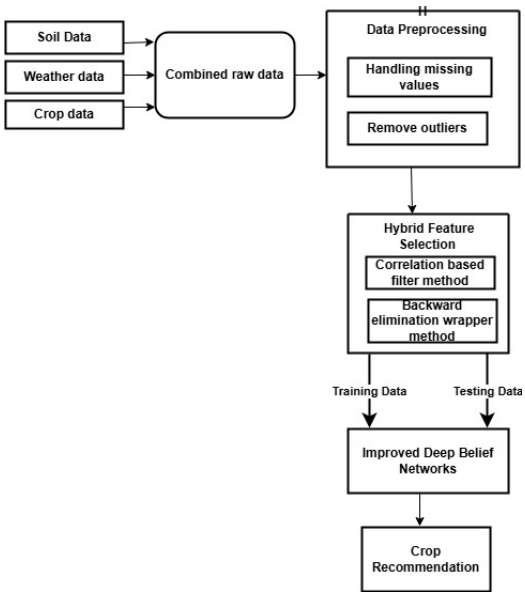

FIGURE 1. Proposed efficient crop recommendation model.

The proposed efficient crop recommendation model consists of two parts. First, the dataset used and the analysis are described. The detailed analysis process is as follows.

Step 1: Efficient features are selected for the input using the hybrid filter and wrapper method.

Step 2: The selected features are input to the visible layer, and the IDBN is pretrained with the Ranger Optimizer to increase the convergence speed considering the weather data from 1972 to 2000 along with soil and crop data.

Step 3: The IDBN was tested with weather data from to 2011-2020 along with soil and crop data, while the data for the period from 2001 to 2010 were considered for validation.

Step 4: The crop recommendation results obtained from the IDBN are compared with those of the DBN, Artificial neural networks, and decision trees.

The algorithmic background of the proposed IDBN approach is described as follows:

A. RESTRICTED BOLTZMANN MACHINE (RBM)

The Restricted Boltzmann Machine (RBM) [33], a variant of the Boltzmann Machine first proposed by Smolensky, generally comprises two key components: a visible layer and multiple hidden layers within the neural network. It is an energy-based model structured as a bipartite graph, as depicted in Figure 2. It comprises a visible layer that serves as the input layer for the data, and a hidden layer. All the visible layer nodes were fully connected to all the hidden layer nodes, whereas there were no connections within each layer. Both the visible and hidden layer nodes are binary variables with values limited to zero or one, respectively.

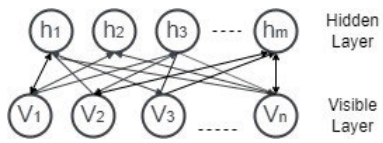

FIGURE 2. Restricted boltzmann machine.

The visible layer accepts the input data, while the important features in the input are identified by the hidden layer through weighted connections. The RBM learning process employs a contrastive divergence training strategy to estimate the relationship between the network's weights and their gradients[36]. The gradient focuses on guiding the learning process towards optimal network parameters. The visible layer is treated as a binary-valued input, for the visible layer and consecutive hidden layers are expressed as $v = \{v_1, v_2, v_3 \ldots v_x\}$ and $h = \{h_1, h_2, h_3 \ldots h_x\}$ respectively. For the hyperparameters $set = \{W, a_v, a_h\}$ the energy function can be expressed [27] as shown in equation1

$$E(h, v \mid \theta) = -\sum_{p=1}^{n} a_{vp} v_p - \sum_{q=1}^{m} a_{hq} h_q - \sum_{p=1}^{n} \sum_{q=1}^{m} v_p h_q w_{pq} \quad (1)$$

where a_v and a_h represent the bias values for the visible and hidden layers, respectively, while w_{pq} denotes the weight linking the neurons in these layers. The probability distribution function for the visible and hidden layers is expressed as shown in Equation 2

$$P(v, h \mid \theta) = \frac{e^{-E(h,v|\theta)}}{\sum_{h,v} e^{-E(h,v|\theta)}} 1001 \text{ [27]} \quad (2)$$

A contrastive divergence algorithm was employed to train the RBM. This process entails providing an input vector for visible units and alternately sampling the hidden units P(h|v)

and visible units P(v|h). The hidden units are independent of each other when conditioned on the visible units, and the same applies in reverse. Under these circumstances, the RBM is known as a Conditional Restricted Boltzmann Machine (CRBM) [33]. The conditional probabilities for the binary-valued hidden and visible units can be calculated using Equation 3:

$$P(h_q = 1 \mid v, \theta) = \varsigma\left(a_{hq} + \sum_{q=1}^{m} w_{pq} v_p\right) \tag{3}$$

With the hidden values, the new values for the visible layer are calculated using equation 4 [27]

$$P(v_p = 1 \mid h, \theta) = \varsigma\left(a_{vp} + \sum_{p=1}^{n} w_{pq} h_q\right) \tag{4}$$

where $\varsigma(z)$ is the sigmoid function given by

$$\varsigma(z) = \frac{1}{1 + e^{-z}} \tag{5}$$

The primary goal of training an RBM is to achieve the optimal hyperparameter $\theta = \{W, a_v, a_h\}$ for the entire network. The process of updating weights and biases is carried out using the Contrastive Divergence (CD) algorithm. The CD algorithm begins by initializing the visible unit state using a training sample and computing the binary state of the hidden units.

B. DEEP BELIEF NETWORKS (DBN)

A deep belief network (DBN) is constructed by stacking RBM layers [38]. Each RBM layer interacts with both the preceding and succeeding layers. A DBN is formed by stacking several RBMs through a simple computation method, which defines the overall structure, except for the final layer where the output is determined using the conditional probability equation as shown in Equation 6:

$$P(v, h_1, h_2, \ldots h_q) = \prod \left(P(h_{q-1} \mid h_q) P(h_{q-1}, h_q)\right) \tag{6}$$

where h_q represents the number of hidden layers. Deep Belief Networks (DBN) composed of multiple two-layer networks, where in each n hidden layer serves as the output for the preceding node and the input for the next node, except in the case of the initial input layer and the final output layer [27]. The DBN employs a layerwise pretraining approach, starting from the lower layers (closer to the input) and moving upward. The training process is conducted in multiple steps, beginning with the first layer of the Restricted Boltzmann Machine (RBM) [33] to establish the training parameters. The hidden layer's output from the first RBM is used as the input [37] for the next RBM layer, and this process is repeated iteratively to train all subsequent layers. Ultimately, the output of the final hidden layer is fed into a softmax classifier, and fine-tuning is conducted using the supervised Gradient Descent (GD) algorithm [27]. The structure of the DBN is shown in Figure 3.

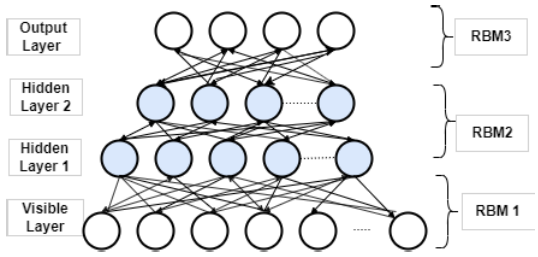

FIGURE 3. Structure of DBN.

C. DEEP BELIEF PROPOSED IMPROVED NETWORKS (IDBN)

The Bernoulli model is employed in the traditional RBM; however, it requires the visible layer to be input with binary values. This can cause loss of information when using continuously distributed input weather data in crop recommendations. To address this issue, the Gaussian Restricted Boltzmann Machine (GRBM) was introduced, which incorporates a Gaussian function to enable non-binary input values to be used without loss of information. The energy function of the GRBM is expressed in equation 7.

$$E(h, v \mid \theta) = -\sum_{p=1}^{n}\left(\frac{(v_p - a_{vp})}{2\sigma^2}\right) - \sum_{q=1}^{m}(a_{hq} h_q) - \sum_{p=1}^{n}\sum_{q=1}^{m}\left(\frac{v_p}{\sigma_p} h_q w_{pq}\right) \tag{7}$$

where n represents the count of visible units in the visible layer,while m denotes the number of hidden units in the hidden layer. The first term represents the energy associated with the visible layer, considering the deviation of observed value of the visible unit v_p from its mean a_{vp} and scaled by the variance σ^2. The second term captures the biases in the hidden layer. The third term describes the interaction energy between the visible layers (v_p) and hidden layers (h_q), scaled by the variance of the visible units. The conditional probabilities of hidden and visible units based on can be computed using the equation 8 and 9 respectively:

$$P(v_p = 1 \mid h, \theta) = N\left(a_{vp} + \sigma_p \sum_{p=1}^{n} h_q w_{pq}, \sigma^2\right) \tag{8}$$

$$P(h_q = 1 \mid v, \theta) = \sigma\left(a_{hq} + \sum_{q=1}^{m} \frac{v_p}{\sigma_p} w_{pq}\right) \tag{9}$$

where $N(\mu, \sigma^2)$ is the Gaussian distribution, where μ is the mean, and σ^2 is the variance.

In equation 8, the mean μ for $P(v_p = 1 \mid h, \theta)$ is $a_{vp} + \sigma_p \sum_{p=1}^{n} h_q w_{pq}$ and the variance σ^2 is variance of the Gaussian distribution. In equation 9 $P(h_q = 1 \mid v, \theta)$ describes the probability of the hidden unit $h_{p,q}$ being active, where represents the sigmoid activation function [14], [37]. The architecture of the IDBN with the GRBM layer is shown in Figure 4. The bottom layer of the IDBN model consists

of a GRBM to handle continuous input data by modeling it using Gaussian distributions, whereas the upper layers are the RBMs. The IDBN model is constructed by stacking GRBM and RBMs in a bottom-up fashion. At the top of the model, a GD algorithm is incorporated to fine-tune the parameters of the IDBN model.

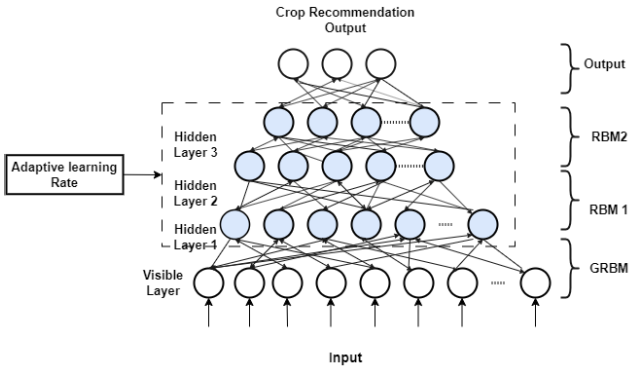

FIGURE 4. Architecture of improved DBN.

Ranger Optimizer

The Ranger optimizer is a cutting-edge optimization algorithm that combines the strengths of Rectified Adam (RAdam) and Lookahead, providing substantial improvements in stability, performance, and convergence speed for deep learning models, including Deep Belief Networks (DBNs) [37]. This optimizer is particularly effective for complex models and datasets, making it an excellent choice for enhancing the performance of DBNs in tasks such as crop recommendation.

Components of Ranger optimizer are: Rectified Adam (RAdam) and Lookahead.

Rectified Adam (RAdam) consists of following steps:

1) **Adaptive Learning Rate:** RAdam rectifies the variance of the adaptive learning rate by adapting it based on the number of training steps. This correction leads to more stable and reliable updates compared to traditional Adam, especially in the early stages of training.
2) **Variance Rectification:** It addresses the variance issue in Adam, where the learning rate can become too large in the initial steps, causing instability. RAdam adjusts the adaptive learning rate to ensure it remains within a stable range.

The outline of adaptive learning rate method is: Define the learning rate range: Determine a range for the learning rate. This range should encompass values both higher and lower than the initial learning rate. For example, range can be set to be in between $\eta_{\min}$ and $\eta_{\max}$:

1) Initialize the learning rate: Set the initial learning rate as the midpoint of the range

$$\eta = \frac{\eta_{\min} + \eta_{\max}}{2}$$

Iterate through each mini-batch during training

2) a) Perform a forward pass to obtain the model's predictions.
   b) Calculate the error between the predictions and the target values.
   c) Update the learning rate η based on the error

An increment factor I and decrement factor D is considered with the learning rate η as shown in equations 10 and 11

$$I\eta^{\text{old}} \text{ if } \left((v_p h_q)_{k-1} - (v_p h_q)_k\right) > 0 \quad (10)$$

$$D\eta^{\text{old}} \text{ if } \left((v_p h_q)_{k-1} - (v_p h_q)_k\right) < 0 \quad (11)$$

The learning rate will be increased by factor I if the previous two outputs are in the same direction. The learning rate will be decreased if the previous two outputs are in a different direction. The values considered in the adaptive learning rate are as follows:

$$\eta_{\min} = 10^{-4}, \quad \eta_{\max} = 0.1, \quad I = 1.1, \quad D = \frac{1}{2}$$

Perform a backward pass (back propagation) to fine tune the model's weights using the adjusted learning rate.

3) Repeat steps 3a to 3d until training is complete.

Lookahead consists of following steps:

a) **Exploration and Exploitation:** Lookahead improves optimization by alternating between "fast" and "slow" weights. Fast weights are frequently updated, while slow weights are updated periodically by looking ahead at the trajectory of fast weights. This mechanism balances exploration and exploitation, leading to more robust convergence.
b) **Stabilization:** By periodically interpolating between fast and slow weights, Lookahead prevents the optimizer from getting stuck in sharp local minima, promoting smoother and more stable convergence.

The Lookahead optimization algorithm enhances the performance of traditional optimizers by alternating between two sets of weights: "fast" and "slow" weights. This method strikes a balance between exploring new regions of the parameter space and refining the existing solution. Here's a detailed explanation with equations.

Notation:

θ_t = Fast weights at step t

$\emptyset_t$ = Slow weights at step t

α = Learning rate for the inner optimizer.

K= Number of steps for the fast weights before updating slow weights.

β= Interpolation factor between fast and slow weights.

Step-by-Step Process:

- Initialize: Set fast weights equal to slow weights: $\theta_0 = \emptyset_0$
- Inner Loop (Fast Weights Update):
  For t from 1 to k:

Update fast weights using Adam optimizer:

$$\theta_t = \theta_{(t-1)} - \alpha \nabla L(\theta_{(t-1)})$$

Here, $\alpha \nabla L(\theta_{(t-1)})$ is the gradient of the loss function L with respect to fast weights θ_{t-1}

- Outer Loop (Slow Weights Update):
  After k steps, update slow weights by interpolating between slow and fast weights: $\emptyset_{(t+1)} = \emptyset_t + \beta(\theta_t - \emptyset_t)$
  Reset fast weights to the updated slow weights: $\theta_{(t+1)} = \emptyset_{(t+1)}$ [27]

Inner Loop (Fast Weights Update): The fast weights θ are regularly updated based on the gradient of the loss function. This enables the optimizer to quickly navigate through the parameter space.

Outer Loop (Slow Weights Update): The slow weights $\emptyset$ are updated less often, with their update determined by interpolating between the current slow weights and the final fast weights after k steps [27]. The interpolation factor β controls the balance between exploration and exploitation. A higher β emphasizes the new information from the fast weights, while a lower β retains more of the old information from the slow weights.

The slower weights gradually synchronize with the faster weights, aiding in the smoothing of the optimization trajectory and preventing sudden changes that could lead to instability. The Lookahead optimizer's alternating updates between fast and slow weights allow it to balance the need for rapid exploration and steady exploitation. The fast weights rapidly explore the parameter space, while the slow weights provide a stabilizing influence by gradually incorporating the fast weights' updates. This leads to more robust convergence, as the optimizer avoids both excessive oscillations and premature convergence to suboptimal solutions.

The Ranger optimizer has improved the convergence time of the crop recommendation model discussed in the later section. By acclimatizing the adaptive learning rate and Lookahead optimizer, the convergence speed of the IDBN model was enhanced compared to the traditional DBN [37], demonstrating the effectiveness of the Ranger optimizer in this application.

IV. DATA AND PREPROCESSING

A. DATASET USED

This study investigates crop recommendations for the state of Karnataka, India. The soil data is collected from the study area is shown in figure 5. The crops considered for recommendation were Rice, Sugarcane, Maize, and Fingermillet. Depending on the crop yield, two districts which correspond to maximum yield for each crop is selected as shown in table 1. For example, the yield of rice is maximum in Bellary and Chikmagalur district for the year 2020, hence the characteristics of these two places are considered. It was ensured that there was no overlap and that the selected area corresponded to one crop. Considering that crops are dependent on soil and weather parameters, the soil characteristics for different survey numbers of the identified areas were acquired from the National Bureau of Soil Survey and Soil Usage Planning (NBSS and LUP) [38]. The survey numbers correspond to individual land holdings. Daily meteorological data from 1972 to 2020 were collected from an open government data platform in India for the identified locations [39].

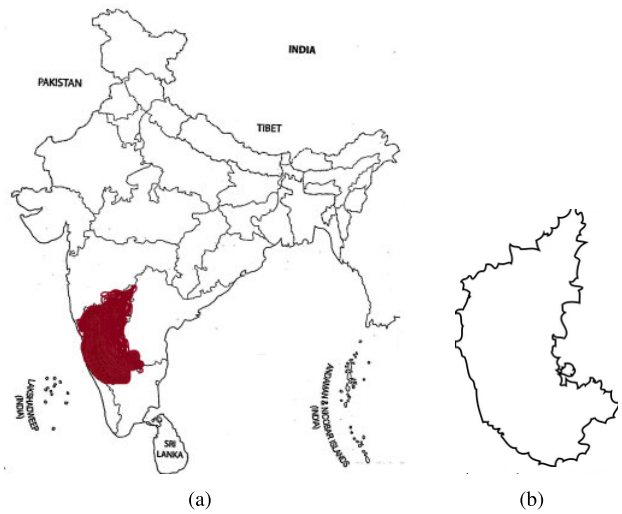

FIGURE 5. Study area.

TABLE 1. Locations considered for every crop.

Crop	Location
Rice	Bellary, Chikmagalur
Sugarcane	Mandya, Haveri
Maize	Belagavi, Vijayapura
Fingermillet	Chikballapur, Tumakuru

Crop production depends on several factors. Crucial factors that influence crop production have been identified. The dataset considered for the experiment consisted of crop, soil, and climate parameters. Crop characteristics include effective root depth, Length of Growing Period (LGP), and evapotranspiration. The physical soil parameters considered included the soil texture, gravel code, Slope, Drainage, Erosion code, and depth. The chemical soil parameters included soil pH, Potassium, Phosphorus, Nitrogen, and Electrical conductivity. The climate factors examined included maximum and minimum temperatures, average temperature, wind speed, humidity, sunshine duration, and potential evapotranspiration [37]. Daily meteorological data for the location from 1972 to 2020 were collected from an open government data platform in India [39]. The soil properties for the designated location were sourced from the National Bureau of Soil Survey and Land Use Planning (NBSS LUP) located in Bengaluru [37].

B. DATASET PREPARATION

The study integrates features from diverse perspectives, such as climate, soil, and crop characteristics, to create a robust

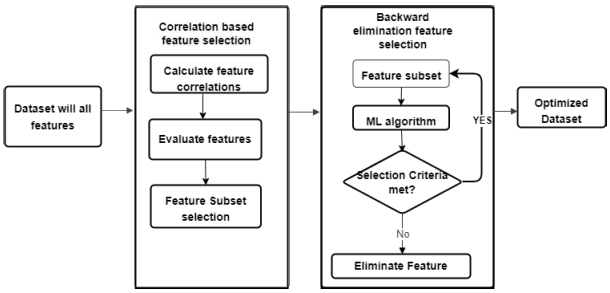

FIGURE 6. Integrated feature selection strategy.

dataset capable of delivering accurate crop recommendations. Feature selection was conducted using a hybrid approach combining Correlation based filter feature selection method and Wrapper Method: Backward-Elimination Feature Selection (BEFS), as illustrated in Figure 6. The subsequent section elaborates on the distinct sections in our proposed hybrid feature selection method. Initially, the combined data undergoes preprocessing, involving the replacement of missing values with the mean or average value specific to each feature. Subsequently, the exclusion of extreme values from the dataset is conducted to mitigate potential inaccuracies arising from data variability. This is succeeded by the essential feature selection process aimed at enhancing accuracy.

The dataset consists of 24 features: Soil texture (St), Soil pH (SpH), Gravel code (Sgc), Erosion code (Sec), Slope (Ssl), Soil drainage (Sdr), Depth(Sdp), Spt(Penetration), oc(Soil organic carbon), av_k(Soil potassium), av_{ni}(Soil nitrogen), av_{fe}(Soil Ferrous), av_{Cu}(Soil Copper), av_{zn}(Soil zinc) av_p(Soil phosphorus), LGP(Length of the growing period), Erd(Effective root depth), mn_{temp}(Minimum temperature), mx_{temp}(Maximum temperature), Hum(Humidity), WS(Wind speed), and SSH(Sunshine hours), RH(Relative humidity). This filter stage resulted in the selection of 21 features, namely: St, pH, Sgc, Ssl, Sec, Sdr, Sdp, Spt, oc, av_{mn}, av_k, av_{ni}, av_{fe}, av_p, LGP, Erd, mn_{temp}, mx_{temp}, Hum, WS, and SSH.

On the other hand, the backward elimination feature selection method is based on the p-value, which is calculated for all the features obtained from the filter method with respect to the dependent variable.

The features selected after this phase resulted in the selection of 19 features, namely St, SpH, Sgc, Sec, Sdr, Sdp, Spt, oc, av_k, av_{mn}, av_{ni}, av_p, ph, LGP, Erd, mx_{temp}, Hum, WS, and SSH. The selected features are input to the visible layer, and the IDBN is pretrained with the Ranger Optimizer to increase the convergence speed.

This research integrated climate data from 1972 to 2000 with soil and crop parameters to construct the training dataset [37]. For the validation dataset, climate data from 2000 to 2010 were paired with corresponding soil and crop parameters, while the test dataset was created by combining climate data from 2010 to 2020 with soil and crop parameters [33].

Similarly, climate data from 2010 to 2020 were combined with soil and crop parameters to form the test dataset [37].

C. DETERMINING PARAMETERS FOR PROPOSED IDBM

Choosing appropriate network parameters is the primary requirement of neural networks. The architecture of the neural network is defined by modifying the number of hidden layers and the number of neurons in each layer [37]. The optimal number of hidden layers should be determined through experimentation, because a lower number of hidden layers may lead to a weaker model, and too many hidden layers may result in overfitting of the model. The number of hidden layers was selected to range from two to six and was investigated using a different number of neurons for each layer.

TABLE 2. Network structure of improved DBN.

# of Hidden Layers	Neurons in Each Hidden Layer	MSE (Validation Data)
2	(9, 14)	0.80
2	(4, 8)	0.64
3	(5, 10, 15)	0.31
3	(6, 11, 14)	0.35
4	(4, 8, 11, 14)	0.25
4	(7, 9, 13, 11)	0.42
5	(8, 7, 14, 12, 15)	0.12
5	(14, 4, 13, 8, 10)	0.19
6	(12, 10, 8, 3, 8, 7)	0.54
6	(15, 14, 7, 5, 11, 4)	0.58

The recommendation results were analyzed for ten different structures to resolve the ideal structure. The structures were evaluated based on the Mean Square Error (MSE) [37] of the validation data, with the results presented in Table 2. Based on the results in the table, it can be concluded that the optimal DBN structure is (8, 7, 14, 12, 15), which implies that the DBN would have five hidden layers, with every layer having 8, 7,14,12,15 neurons respectively from the 1st to the 5th layer. The output layer has four neurons, where each neuron indicates the suitability level of the crops, namely rice, sugarcane, maize, and finger millet. The DBN parameter settings for the proposed method are listed in Table 3. The proposed improved DBN algorithm with an adaptive learning step was compared with three other algorithms: DBN, Artificial Neural Networks (ANN), and Decision Tree.

TABLE 3. Parameter setting for improved DBN.

Parameters	Details
Forecasting Variable	Location-specific crop recommendation classes
Input Variable	Soil characters, Climate characters and Crop Characters
Training Parameters	Learning Rate: Determined by adaptive learning rate Batch Size: 20 No of Hidden Units: 5 hidden layers with 8, 7, 14, 12, and 15 neurons respectively

D. EVALUATION MEASURES

The evaluation measures used to assess multiclass classification are as follows: The classifier's performance was

assessed using True Positive (TP), TrueNegative (TN), False Positive (FP), and False Negative (FN) values derived from the test data [39]. The classification accuracy of the model was determined by defining the number of correct predictions among all predictions [40] as shown in equation 12. Accuracy enables us to determine effective results for every class.

$$\text{Accuracy} = \frac{\text{TP}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{12}$$

Measures, such as precision, recall, and F1 scores [27], were generalized for multiclass classification. In multiclass classification, Micro- and macro-averaging methods can be used in multiclass classification. The macro average calculates the metrics for each class separately and then averages them, giving equal weight to all classes [37]. Micro averaging considers the contributions of all classes by aggregating the measures, that is, precision, recall, and F1 score, which were obtained for all classes together. Precision is the ratio of correct prediction that are actually true as shown in equation 13 and 14.

$$\text{Precision}_{\text{micro}} = \frac{\sum_{p=1}^{n} TP_p}{\sum_{p=1}^{n}(FP_p + TP_p)} \tag{13}$$

$$\text{Precision}_{\text{macro}} = \frac{\sum_{p=1}^{n} \frac{TP_i}{FP_p + TP_p}}{n} \tag{14}$$

Recall is the proportion of correctly predicted positive instances out of all actual positives, as shown in equations 15 and 16.

$$\text{Recall}_{\text{micro}} = \frac{\sum_{p=1}^{n} TP_p}{\sum_{p=1}^{n}(FN_p + TP_p)} \tag{15}$$

$$\text{Recall}_{\text{macro}} = \frac{\sum_{p=1}^{n} \frac{TP_p}{FN_p + TP_p}}{n} \tag{16}$$

The F1 score is calculated as the harmonic mean of precision and recall, as expressed in the equation 17 and 18.

$$F1Score_{\text{micro}} = 2 * \frac{\text{precision}_{\text{micro}} * \text{recall}_{\text{micro}}}{\text{precision}_{\text{micro}} + \text{recall}_{\text{micro}}} \tag{17}$$

$$F1Score_{\text{macro}} = 2 * \frac{\text{precision}_{\text{macro}} * \text{recall}_{\text{macro}}}{\text{precision}_{\text{macro}} + \text{recall}_{\text{macro}}} \tag{18}$$

Errors were evaluated using micro- and macro-averaging, along with metrics such as mean squared error (MSE), mean absolute error (MAE), and root mean squared error (RMSE). The calculations are shown in equations 19, 20, and 21:

$$\text{MSE} = \frac{1}{n}\sum_{p=1}^{n}(P_p - A_p)^2 \tag{19}$$

$$\text{MAE} = \frac{1}{n}\sum_{p=1}^{n}|P_p - A_p| \tag{20}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{p=1}^{n}(P_p - A_p)^2}{N}} \tag{21}$$

where P represents the predicted value and A denotes the actual value. Log loss for the neural networks is calculated as shown in equation 22.

$$\text{Log Loss} = -\frac{1}{n}\sum_{p=1}^{n}\left[A_p \log(P_p) + (1 - A_p)\log(1 - P_p)\right] \tag{22}$$

V. EVALUATION AND DISCUSSION OF RESULTS

This section outlines the experimental outcomes obtained using the proposed enhanced Deep Belief Networks. The proposed method's performance was evaluated against other machine learning algorithms. Error metrics were computed for the test dataset, and the results are presented in Table 4. These results obtained unequivocally show that the IDBN outperforms the other machine learning algorithms evaluated. The adaptive learning rate in the IDBN finds the optimal learning rate for every iteration, ensuring that the minimum error values are reached in 164 iterations compared with the conventional DBN. The stopping criterion considered was 200 iterations or an MSE of 0.1. Table 5 shows the minimum MSE values obtained after 200 iterations for the DBN and ANN, whereas IDBN attained the minimum MSE after 164 iterations.

TABLE 4. Evaluation metrics of different algorithms.

Algorithm	RMSE	MAE	MSE	Accuracy	Log Loss
Improved DBN	0.19	0.18	0.10	93.78%	0.039
DBN	0.22	0.25	0.21	89.47%	0.078
ANN	0.37	0.29	0.32	87.66%	0.114
Decision tree	0.58	0.54	0.50	79.45%	NA

TABLE 5. MSE values of different algorithms.

Algorithm	Iterations	MSE
Improved DBN	164	0.10
DBN	200	0.21
ANN	200	0.32

Weather data from 2011to 2020 were used to measure the accuracy of the recommendation results. An improved DBN model for the recommendation of rice, sugarcane, maize, and finger millet crops. The accuracy of the experimental results obtained was compared with that of the DBN, ANN and Decision Tree, as shown in figure 7. It can be seen from the accuracy curves that Improved DBN outperforms the crop recommendation results by other machine learning algorithms.

A. RECEIVER OPERATING CURVES (ROC)

Receiver Operating Curves (ROC) are evaluation metrics that measures the performance of classification models. The ROC curve represents the probability curve. This indicates how well the model can distinguish between classes. ROC curves graphically depict the relationship between the true positive

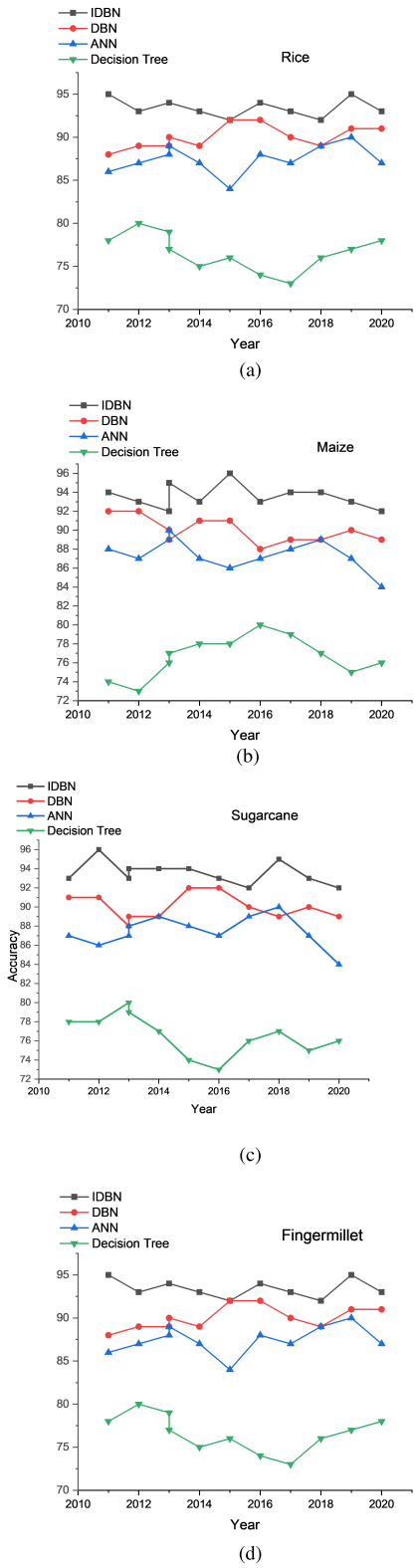

FIGURE 7. Comparison of crop recommendation accuracy (a) Rice crop recommendation (b) Maize crop recommendation (c) Sugarcane crop recommendation (d) Fingermillet crop recommendation.

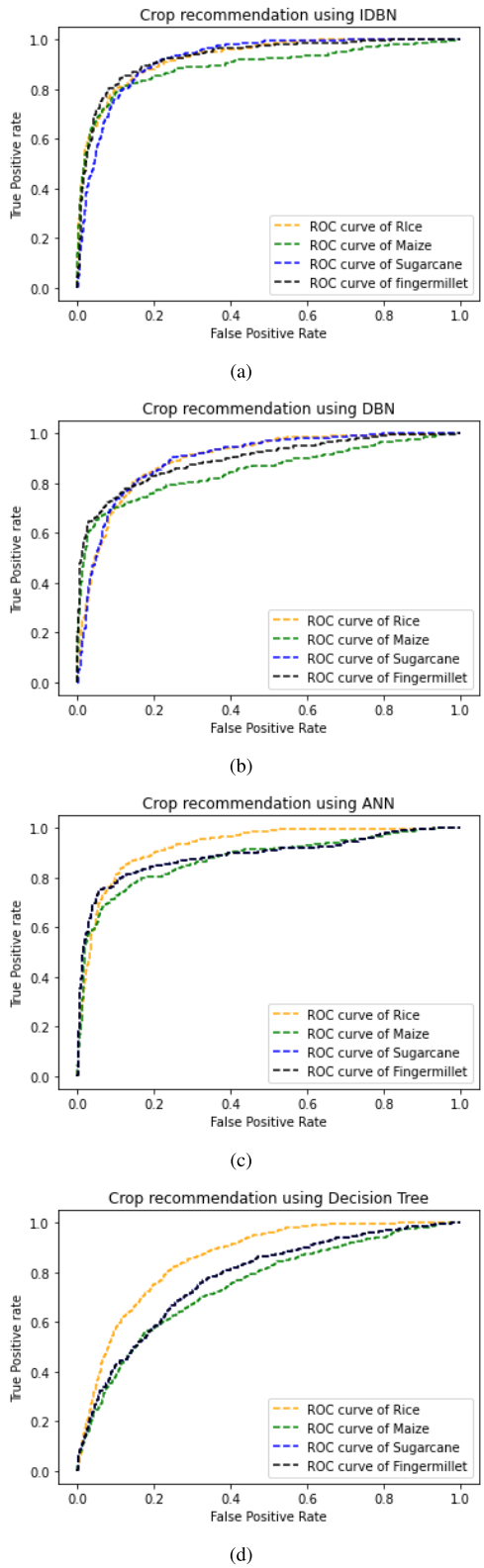

FIGURE 8. ROC curves for (a): Improved DBN (b) DBN (c) ANN (d) Decision Tree.

rate and the false positive rate. ROC curves specify how well positive class probabilities are distinguished from negative probabilities [41]. The ROC curves for the Improved DBN, DBN, ANN, and decision tree are shown in figure 8.

TABLE 6. Comparison of machine learning model performance for Rice and Sugarcane.

Evaluation Metrics	Rice				Sugarcane			
	IDBN	DBN	ANN	Decision Tree	IDBN	DBN	ANN	Decision Tree
Accuracy	0.91	0.90	0.88	0.76	0.92	0.90	0.82	0.71
Precision_micro	0.94	0.87	0.79	0.79	0.87	0.91	0.77	0.69
Precision_macro	0.20	0.22	0.21	0.22	0.22	0.21	0.21	0.22
Recall_micro	0.93	0.85	0.75	0.74	0.85	0.92	0.81	0.68
Recall_macro	0.20	0.25	0.22	0.23	0.25	0.24	0.23	0.21
F1 Score_micro	0.93	0.86	0.76	0.76	0.86	0.92	0.79	0.69
F1 Score_macro	0.20	0.24	0.21	0.28	0.24	0.22	0.22	0.21
AUC-ROC_Score	0.95	0.92	0.84	0.78	0.92	0.89	0.79	0.67

TABLE 7. Performance comparison of the machine learning models for maize and finger millet.

Evaluation Metrics	Maize				Finger Millet			
	IDBN	DBN	ANN	Decision Tree	IDBN	DBN	ANN	Decision Tree
Accuracy	0.91	0.87	0.87	0.69	0.92	0.90	0.87	0.72
Precision_micro	0.92	0.88	0.84	0.707	0.92	0.86	0.81	0.73
Precision_macro	0.22	0.23	0.22	0.24	0.21	0.21	0.20	0.23
Recall_micro	0.89	0.86	0.84	0.724	0.93	0.87	0.79	0.73
Recall_macro	0.23	0.25	0.234	0.22	0.22	0.23	0.21	0.22
F1 Score_micro	0.90	0.87	0.84	0.71	0.91	0.85	0.78	0.74
F1 Score_macro	0.22	0.24	0.228	0.23	0.23	0.23	0.21	0.23
AUC-ROC_Score	0.91	0.861	0.81	0.69	0.91	0.87	0.80	0.72

These models were executed using Python under analogous hardware and software conditions to ensure valid experimental results. The performance of the IDBN-based recommendation model was compared with that of other models. All the results are represented in table 6 and 7. We considered four crops for the recommendation: rice, maize, sugarcane (SC), and fingermillet (FM). The experiments were performed on an IDBN with five hidden layers and eight, seven, 14, 12, and 15 neurons at each hidden layer. The learning rate was improved in every iteration by using an adaptive learning step. Examining the results, the efficacy of the crop-recommendation model based on the IDBN can be compared with other machine-learning algorithms. The error metrics of the MSE, RMSE, and MAE were lower for the IDBN model, as shown in Table 4. The accuracy measure of IDBN is superior to that of other recommendation models, with an accuracy of 94%, as shown in Table 4 and 5. Figure 7 provides a detailed comparison of the IDBN and other machine learning models for each crop. The IDBN achieves an average Area Under the ROC Curve (AUC) score of 0.94, highlighting its effectiveness in capturing dataset variations. Considering the evaluation metrics and ROC curves witnesses that the recommendation model based on IDBN outperforms the other machine learning models. Table 6 and 7 provide a detailed comparison of the IDBN and other machine learning models for individual crops. Though the evaluation results of DBN is similar to IDBN, the convergence time of IDBN is reduced. The soil and climate data are considered from the locations producing the highest yield for the respective crops, the crop recommendation model results certainly help the farmers to make the correct choice.

VI. CONCLUSION

Agriculture is the most significant sector and source of livelihood for many people worldwide. Recommending a suitable crop suiting the location-specific climatic and soil parameters would enhance the final yield lessening the burden on the farmer and available natural resources. This research proposes a crop recommendation model that integrates Enhanced Deep Belief Networks (IDBN) with a Gaussian Restricted Boltzmann Machine (GRBM) and employs the Ranger optimizer for better efficiency. The GRBM enables continuous value inputs to the IDBN, and the adaptive learning rate and Lookahead optimization of Ranger optimizer supports a faster convergence of the optimal results. The evaluation results indicate that the proposed IDBN model outperforms previous machine learning models in terms of performance. Limitation of the study is that only long term food grains are considered. The current study can be enhanced by considering short term crops and vegetables for future recommendations. The work can be extended for crop rotation methods to sustain soil fertility efficiently. As the inclusion of more locations and crops makes the model data-intensive, Big Data platforms must be utilized to enable effective data storage and retrieval.

REFERENCES

[1] J. P. Aryal, T. B. Sapkota, R. Khurana, A. Khatri-Chhetri, D. B. Rahut, and M. L. Jat, "Climate change and agriculture in South Asia: Adaptation options in smallholder production systems," *Environ., Develop. Sustainability*, vol. 22, no. 6, pp. 5045–5075, Aug. 2020.

[2] N. El-Bendary, E. Elhariri, M. Hazman, S. M. Saleh, and A. E. Hassanien, "Cultivation-time recommender system based on climatic conditions for newly reclaimed lands in Egypt," *Proc. Comput. Sci.*, vol. 96, pp. 110–119, Jan. 2016.

[3] S. P. Wani, K. H. Anantha, and K. K. Garg, "Soil properties, crop yield, and economics under integrated crop management practices in Karnataka, Southern India," *World Develop.*, vol. 93, pp. 43–61, May 2017.

[4] L. Zhang, Y. Huang, L. Rong, X. Duan, R. Zhang, Y. Li, and J. Guan, "Effect of soil erosion depth on crop yield based on topsoil removal method: A meta-analysis," *Agronomy Sustain. Develop.*, vol. 41, no. 5, pp. 1–13, Oct. 2021.

[5] R. Amundson, A. A. Berhe, J. W. Hopmans, C. Olson, A. E. Sztein, and D. L. Sparks, "Soil and human security in the 21st century," *Science*, vol. 348, no. 6235, May 2015, Art. no. 1261071.

[6] C. J. Ransom, N. R. Kitchen, J. J. Camberato, P. R. Carter, R. B. Ferguson, F. G. Fernández, D. W. Franzen, C. A. M. Laboski, D. B. Myers, E. D. Nafziger, J. E. Sawyer, and J. F. Shanahan, "Statistical and machine learning methods evaluated for incorporating soil and weather into corn nitrogen recommendations," *Comput. Electron. Agricult.*, vol. 164, Sep. 2019, Art. no. 104872.

[7] P. Liu, Y. Long, H.-C. Song, and Y.-D. He, "Investment decision and coordination of green agri-food supply chain considering information service based on blockchain and big data," *J. Cleaner Prod.*, vol. 277, Dec. 2020, Art. no. 123646.

[8] K. G. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, "Machine learning in agriculture: A review," *Sensors*, vol. 18, no. 8, p. 2674, Aug. 2018.

[9] R. H. L. Ip, L.-M. Ang, K. P. Seng, J. C. Broster, and J. E. Pratley, "Big data and machine learning for crop protection," *Comput. Electron. Agricult.*, vol. 151, pp. 376–383, Aug. 2018.

[10] T. A. Russo, N. Devineni, and U. Lall, "Assessment of agricultural water management in Punjab, India, using Bayesian methods," in *Sustainability of Integrated Water Resources Management*, S. Setegn and M. Donoso, Eds., Cham, Switzerland: Springer, 2015, pp. 147–162, doi: 10.1007/978-3-319-12194-9_9.

[11] Z. Bai, T. Caspari, M. R. Gonzalez, N. H. Batjes, P. Mäder, E. K. Bünemann, R. de Goede, L. Brussaard, M. Xu, C. S. S. Ferreira, E. Reintam, H. Fan, R. Mihelič, M. Glavan, and Z. Tóth, "Effects of agricultural management practices on soil quality: A review of long-term experiments for Europe and China," *Agricult., Ecosystems Environ.*, vol. 265, pp. 1–7, Oct. 2018.

[12] S. S. Kamble, A. Gunasekaran, and S. A. Gawankar, "Achieving sustainable performance in a data-driven agriculture supply chain: A review for research and applications," *Int. J. Prod. Econ.*, vol. 219, pp. 179–194, Jan. 2020.

[13] F.-H. Tseng, H.-H. Cho, and H.-T. Wu, "Applying big data for intelligent agriculture-based crop selection analysis," *IEEE Access*, vol. 7, pp. 116965–116974, 2019.

[14] P. Jayashankar, W. J. Johnston, S. Nilakanta, and R. Burres, "Co-creation of value-in-use through big data technology-a B2B agricultural perspective," *J. Bus. Ind. Marketing*, vol. 35, no. 3, pp. 508–523, Sep. 2019.

[15] D. Elavarasan, D. R. Vincent, V. Sharma, A. Y. Zomaya, and K. Srinivasan, "Forecasting yield by integrating agrarian factors and machine learning models: A survey," *Comput. Electron. Agricult.*, vol. 155, pp. 257–282, Dec. 2018.

[16] T. van Klompenburg, A. Kassahun, and C. Catal, "Crop yield prediction using machine learning: A systematic literature review," *Comput. Electron. Agricult.*, vol. 177, Oct. 2020, Art. no. 105709.

[17] K. Dokic, L. Blaskovic, and D. Mandusic, "From machine learning to deep learning in agriculture—The quantitative review of trends," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 614, no. 1, Dec. 2020, Art. no. 012138.

[18] G. E. Hinton, "A practical guide to training restricted Boltzmann machines," *Momentum*, vol. 9, no. 1, p. 926, Jan. 2010.

[19] J. Luo, C. Zhao, Q. Chen, and G. Li, "Using deep belief network to construct the agricultural information system based on Internet of Things," *J. Supercomput.*, vol. 78, no. 1, pp. 379–405, Jan. 2022.

[20] S. Jana, S. Jana, and A. Begum, "Design and analysis of pepper leaf disease detection using deep belief network," *Eur. J. Mol. Clin. Med.*, vol. 7, no. 9, pp. 1724–1731, Dec. 2020.

[21] A. Gonzalez-Sanchez, J. Frausto-Solis, and W. Ojeda-Bustamante, "Predictive ability of machine learning methods for massive crop yield prediction," *Spanish J. Agricult. Res.*, vol. 12, no. 2, pp. 313–328, Apr. 2014.

[22] Q. Tan, Y. Liu, T. Pan, X. Song, and X. Li, "Changes and determining factors of crop evapotranspiration derived from satellite-based dual crop coefficients in north China plain," *Crop J.*, vol. 10, no. 5, pp. 1496–1506, Oct. 2022.

[23] S. Saleem, M. I. Sharif, M. I. Sharif, M. Z. Sajid, and F. Marinello, "Comparison of deep learning models for multi-crop leaf disease detection with enhanced vegetative feature isolation and definition of a new hybrid architecture," *Agronomy*, vol. 14, no. 10, p. 2230, Sep. 2024.

[24] A. Kumar, S. Sarkar, and C. Pradhan, "Recommendation system for crop identification and pest control technique in agriculture," in *Proc. Int. Conf. Commun. Signal Process. (ICCSP)*, Apr. 2019, pp. 0185–0189.

[25] S. Pudumalar, E. Ramanujam, R. H. Rajashree, C. Kavya, T. Kiruthika, and J. Nisha, "Crop recommendation system for precision agriculture," in *Proc. 8th Int. Conf. Adv. Comput. (ICoAC)*, Jan. 2017, pp. 32–36.

[26] E. R. Abraham, J. G. Mendes dos Reis, O. Vendrametto, P. L. D. Oliveira Costa Neto, R. Carlo Toloi, A. E. D. Souza, and M. D. Oliveira Morais, "Time series prediction with artificial neural networks: An analysis using Brazilian soybean production," *Agriculture*, vol. 10, no. 10, p. 475, Oct. 2020.

[27] J. Madhuri and M. Indiramma, "Artificial neural networks based integrated crop recommendation system using soil and climatic parameters," *Indian J. Sci. Technol.*, vol. 14, no. 19, pp. 1587–1597, May 2021.

[28] S. H. Haji and A. M. Abdulazeez, "Comparison of optimization techniques based on gradient descent algorithm: A review," *PalArch's J. Archaeol. Egypt/Egyptol.*, vol. 18, no. 4, pp. 2715–2743, 2021.

[29] Y. Bengio, *Learning Deep Architectures for AI*. Now, 2009, doi: 10.1561/2200000006.

[30] A. S. Terliksiz and D. T. Altýlar, "Use of deep neural networks for crop yield prediction: A case study of soybean yield in Lauderdale county, Alabama, USA," in *Proc. 8th Int. Conf. Agro-GeoInform.*, Jul. 2019, pp. 1–4.

[31] R. Llugsi, S. E. Yacoubi, A. Fontaine, and P. Lupera, "Comparison between Adam, AdaMax and Adam W optimizers to implement a weather forecast based on neural networks for the Andean city of Quito," in *Proc. IEEE 5th Ecuador Tech. Chapters Meeting (ETCM)*, Oct. 2021, pp. 1–6.

[32] K. Shaukat, S. Luo, S. Chen, and D. Liu, "Cyber threat detection using machine learning techniques: A performance evaluation perspective," in *Proc. Int. Conf. Cyber Warfare Secur. (ICCWS)*, Oct. 2020, pp. 1–6.

[33] N. Zhang, S. Ding, J. Zhang, and Y. Xue, "An overview on restricted Boltzmann machines," *Neurocomputing*, vol. 275, pp. 1186–1199, Jan. 2018.

[34] J. Chen, Q. Jin, and J. Chao, "Design of deep belief networks for short-term prediction of drought index using data in the Huaihe river basin," *Math. Problems Eng.*, vol. 2012, no. 1, Jan. 2012, Art. no. 235929.

[35] G. E. Hinton, "Training products of experts by minimizing contrastive divergence," *Neural Comput.*, vol. 14, no. 8, pp. 1771–1800, Aug. 2002.

[36] S. Hu, Y. Xiang, D. Huo, S. Jawad, and J. Liu, "An improved deep belief network based hybrid forecasting method for wind power," *Energy*, vol. 224, Jun. 2021, Art. no. 120185.

[37] J. Madhuri and M. Indiramma, "Hybrid filter and wrapper methods based feature selection for crop recommendation," in *Proc. Int. Conf. Electron. Syst. Intell. Comput. (ICESIC)*, Apr. 2022, pp. 247–252.

[38] *Bhoomi NBSS&LUP Geo Portal*. Accessed: Aug. 1, 2021. [Online]. Available: https://www.nbsslup.in/bhoomi/

[39] *Open Government Data (OGD) Platform India*. Accessed: Aug. 1, 2021. [Online]. Available: https://data.gov.in/

[40] M. Sokolova and G. Lapalme, "A systematic analysis of performance measures for classification tasks," *Inf. Process. Manage.*, vol. 45, no. 4, pp. 427–437, Jul. 2009.

[41] P. Gholizadeh, B. Esmaeili, and B. Memarian, "Evaluating the performance of machine learning algorithms on construction accidents: An application of ROC curves," in *Proc. Construct. Res. Congr.*, Mar. 2018, pp. 8–18.

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